

AI-Powered Tropical Cyclone Prediction

Final Model Training & Evaluation Report

Team: Hexa-Minds **SIH Problem:** AI/ML-based identification, classification and prediction of tropical cyclone patterns using multi-source satellite data.

Purpose of this report: Document exactly what the final training pipeline did, what data were used, how the LSTM models were trained, what they predict, and how the final test performance should be interpreted.

1. Executive Summary

The final pipeline successfully trained two recurrent neural-network models: an LSTM regression model for 6-hour-ahead cyclone position, wind speed and central pressure, and an LSTM classification model for three intensity classes: Weak, Moderate and Strong. The run used two Tesla T4 GPUs detected by TensorFlow. The final output reports 57,841 cleaned base rows from 1,858 cyclones, with 9,836 complete modeling rows from 340 cyclones.

Headline test results: intensity classification accuracy = **89.57%**; mean 6-hour track error = **95.53 km**; R^2 = 0.992935 for latitude, 0.992727 for longitude, 0.952996 for wind speed and 0.948783 for pressure.

2. Data Used and What Was Actually Trained

The notebook inspected several datasets, including IBTrACS and an Asian Typhoon Dataset. However, the final trained pipeline shown in the output rebuilds/uses the **IBTrACS North Indian Ocean (NI)** records as its modeling base. Therefore, this particular trained result should be described as an IBTrACS-based model, not yet as a fully fused multi-source satellite model. The multi-source fusion remains the broader SIH system objective.

Item	Final output
Base rows	57,841
Cyclones	1,858
Date range	1842-10-25 03:00 to 2025-12-02 18:00
Complete modeling rows	9,836
Complete modeling cyclones	340
Training cyclones / rows	238 / 6,886
Validation cyclones / rows	51 / 1,535
Test cyclones / rows	51 / 1,415

The cyclone-wise split is important: whole cyclone tracks are assigned to one split, reducing leakage from the same storm appearing in both training and testing.

3. Input Features

The final code defines **17 input features**. They combine the cyclone's current state, motion, recent changes and time information:

1	LAT
2	LON
3	WIND_KTS
4	PRESSURE_HPA
5	STORM_SPEED

6	STORM_DIR_SIN
7	STORM_DIR_COS
8	WIND_CHANGE
9	PRESSURE_CHANGE
10	LAT_CHANGE
11	LON_CHANGE
12	DIST2LAND
13	MONTH_SIN
14	MONTH_COS
15	HOUR_SIN
16	HOUR_COS
17	YEAR

The sequence length is **6 consecutive observations at 3-hour intervals**. Thus the model receives 18 hours of recent cyclone history (6 observations × 3 hours between observations) and predicts the next 6-hour state.

4. What the Regression Model Predicts

The regression LSTM has four outputs for the next 6 hours:

Output	Meaning
NEXT_6H_LAT	Predicted cyclone latitude after 6 hours
NEXT_6H_LON	Predicted cyclone longitude after 6 hours
NEXT_6H_WIND_KTS	Predicted wind speed after 6 hours
NEXT_6H_PRESSURE_HPA	Predicted central pressure after 6 hours

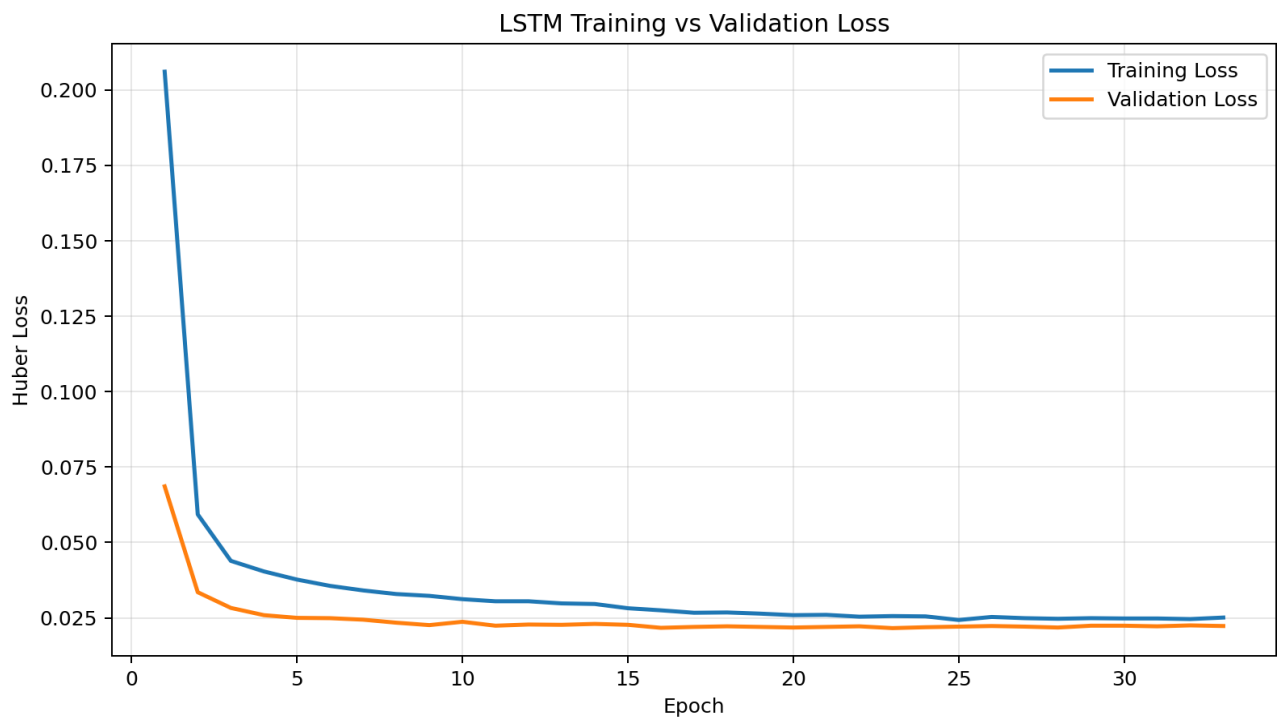
This means the model is not simply answering whether a cyclone exists. It estimates **where the cyclone will be, how fast/strong it will be, and what its pressure will be** 6 hours into the future.

5. Regression LSTM Architecture

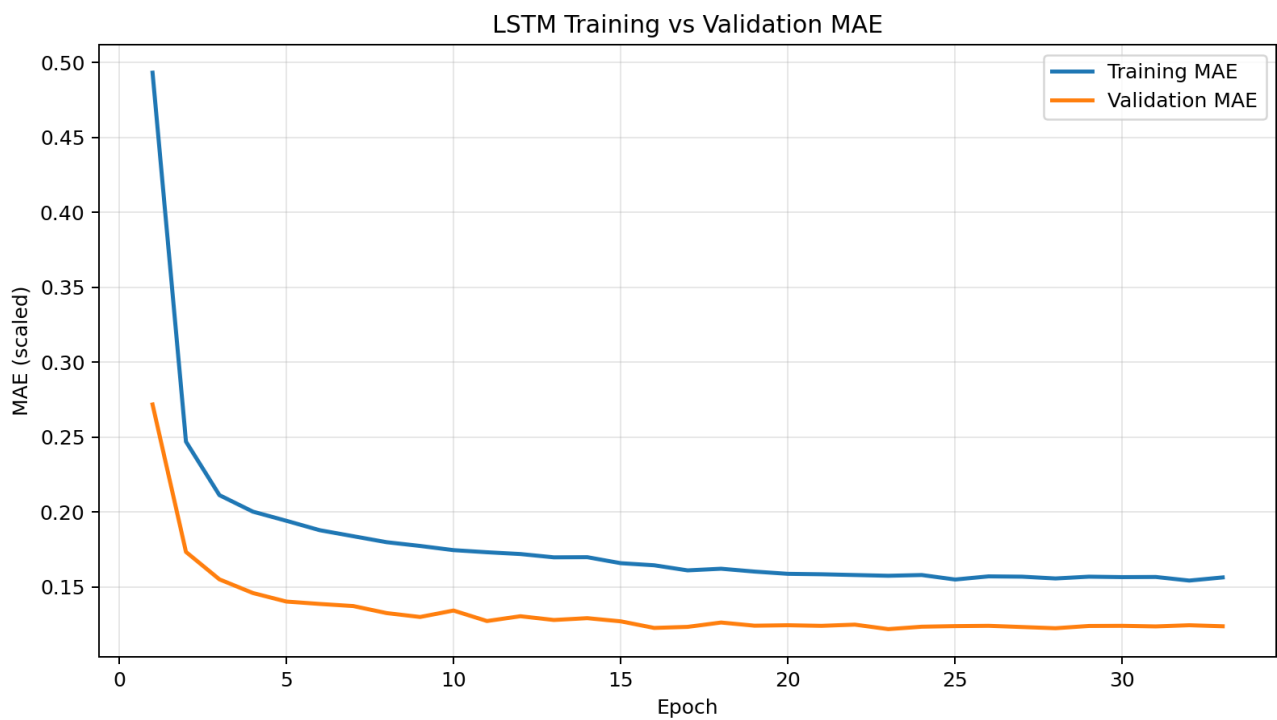
Layer	Configuration
Input	6 time steps × 17 features
LSTM 1	128 units, return sequences = True
Dropout	0.20
LSTM 2	64 units
Dropout	0.20
Dense	64 units, ReLU
Dropout	0.10
Dense	32 units, ReLU
Output	4 neurons
Parameters	130,532 trainable
Optimizer	Adam, learning rate 5×10^{-4}
Loss	Huber loss
Metric	MAE

The pipeline also uses ModelCheckpoint to keep the best validation model, ReduceLROnPlateau to lower the learning rate when validation loss stops improving, and EarlyStopping with patience 10.

6. Regression Training Behaviour



The best validation loss occurred at epoch 23. Training continued to epoch 33, then early stopping restored the epoch-23 weights. This is why the run did not need to use all 60 requested epochs.



Validation MAE reached approximately 0.1218 in the scaled training space at epoch 23. The decreasing curves and callback behaviour show that the model learned and then reached a plateau.

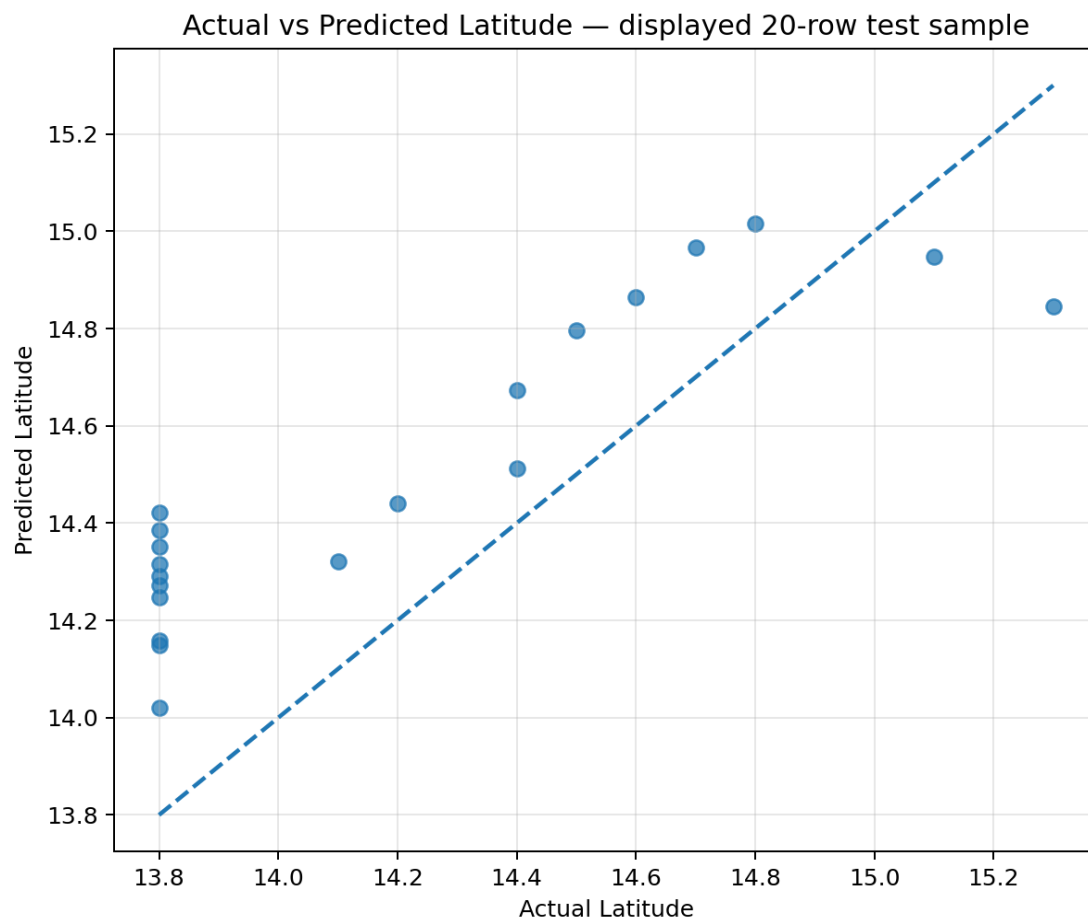
7. Final Regression Test Metrics

Target	MAE	RMSE	R ²
Next 6h latitude	0.301791°	0.392106°	0.992935
Next 6h longitude	0.777185°	0.982971°	0.992727
Next 6h wind speed	3.621397 kt	5.395151 kt	0.952996
Next 6h pressure	2.328919 hPa	3.655259 hPa	0.948783

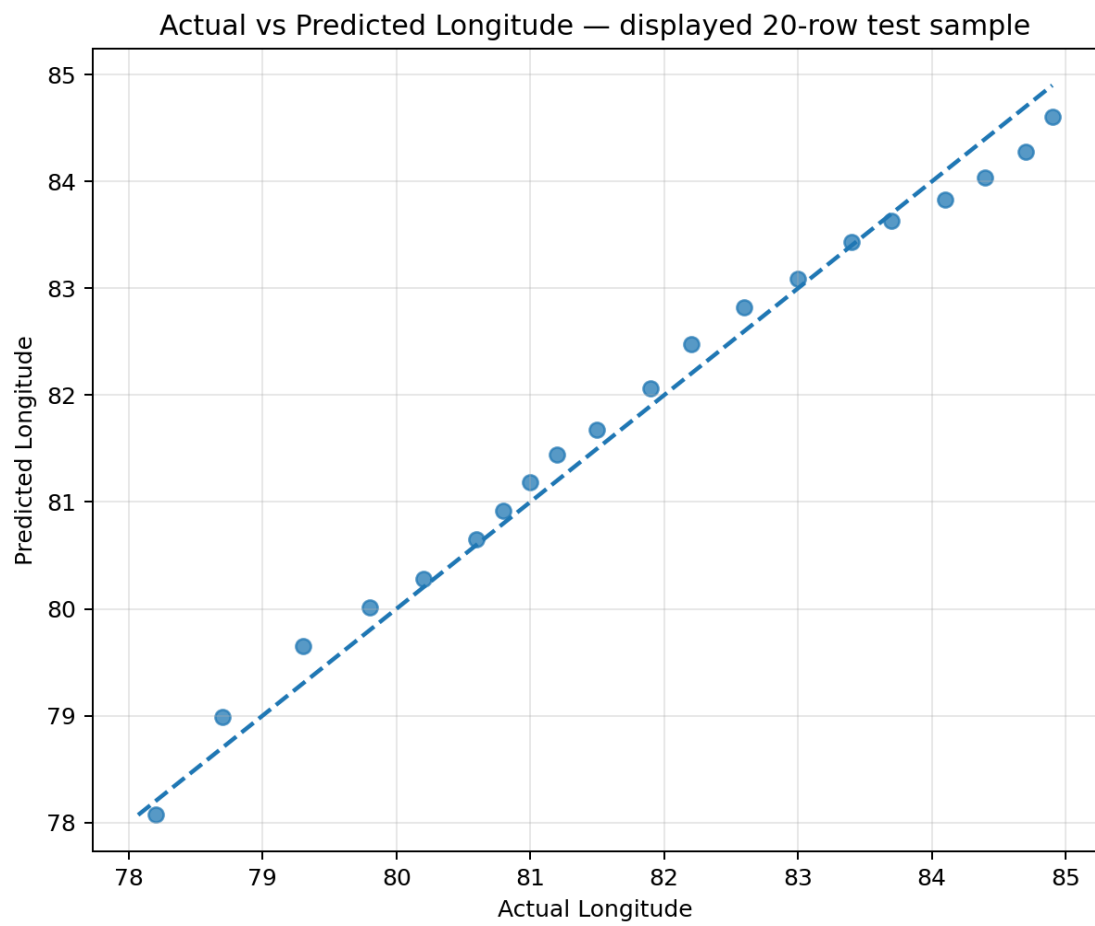
Interpretation: latitude and longitude have R² values around 0.993, indicating that the predicted 6-hour positions explain most of the variance in the held-out test targets. Wind and pressure also show strong fit, with R² values above 0.94.

Important: R² is not an accuracy percentage. Report the regression results using MAE, RMSE, R² and physical track error.

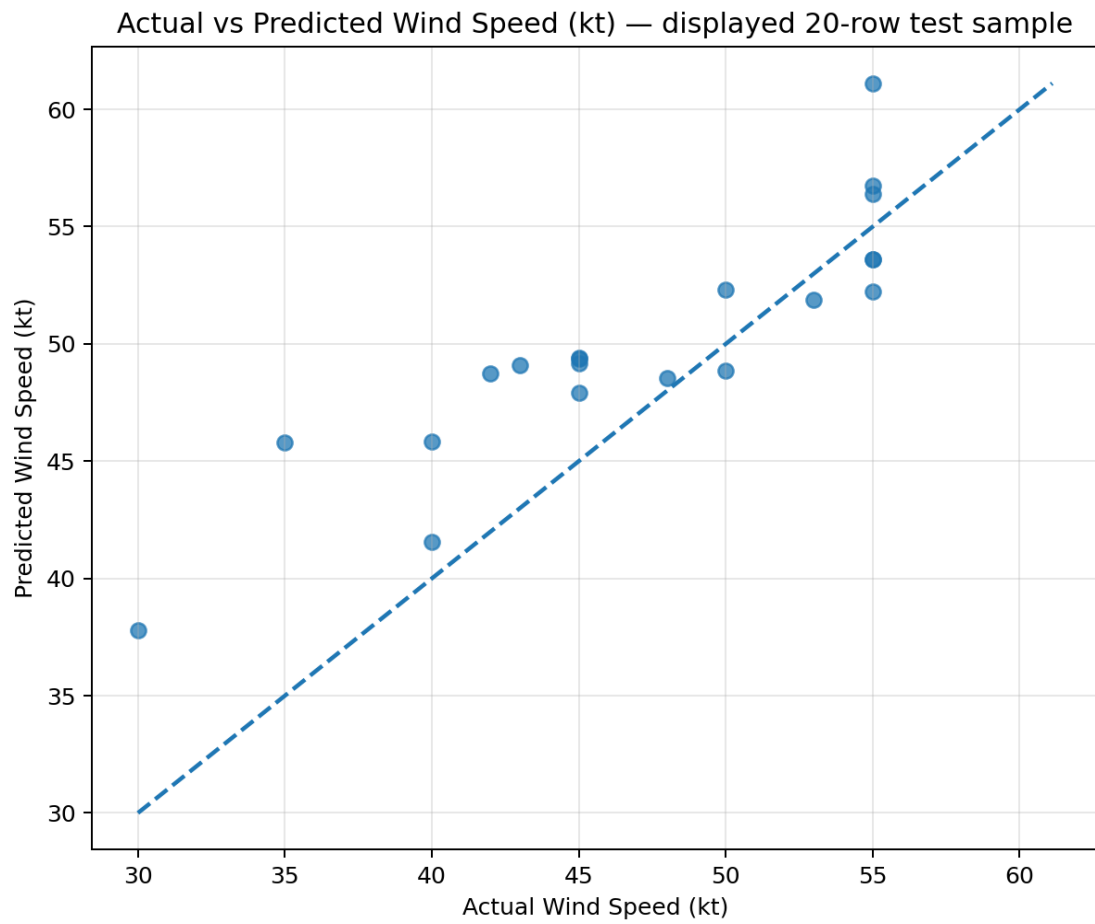
8. Actual vs Predicted — Latitude



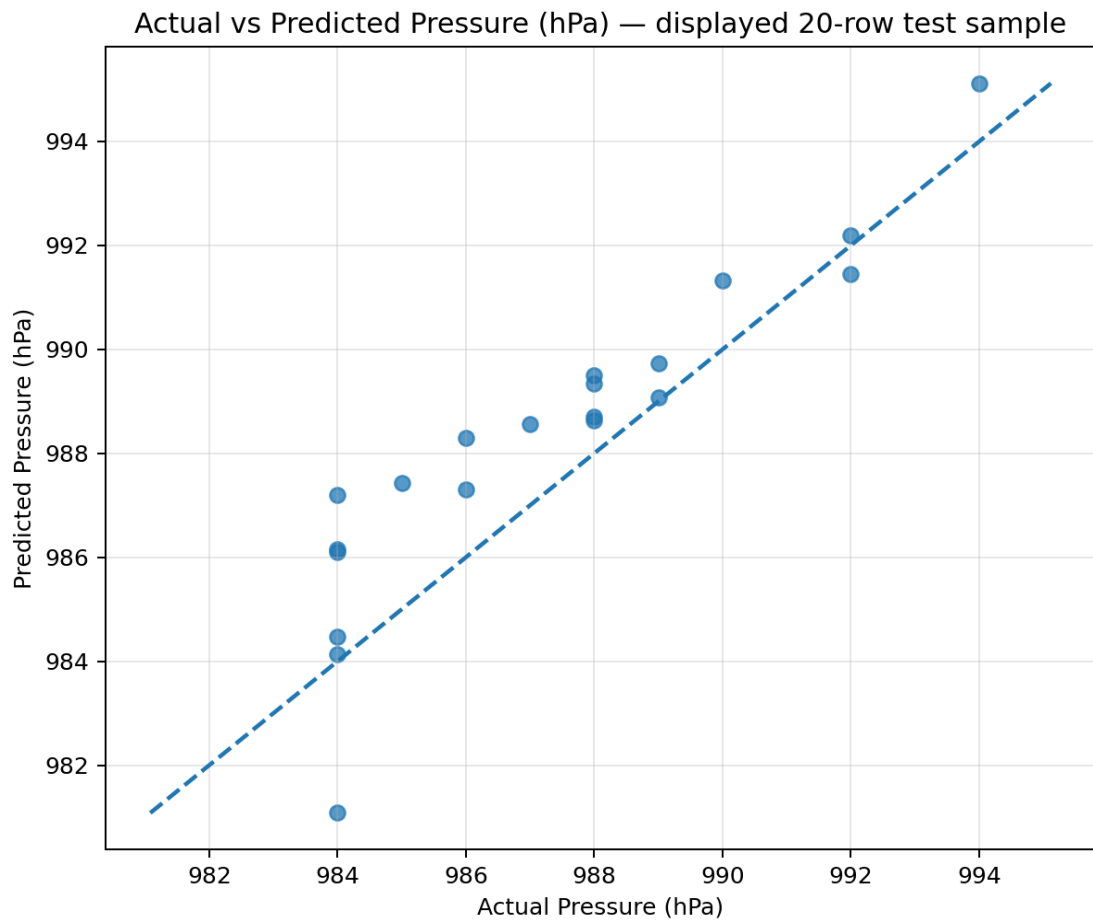
8. Actual vs Predicted — Longitude



8. Actual vs Predicted — Wind Speed (kt)



8. Actual vs Predicted — Pressure (hPa)

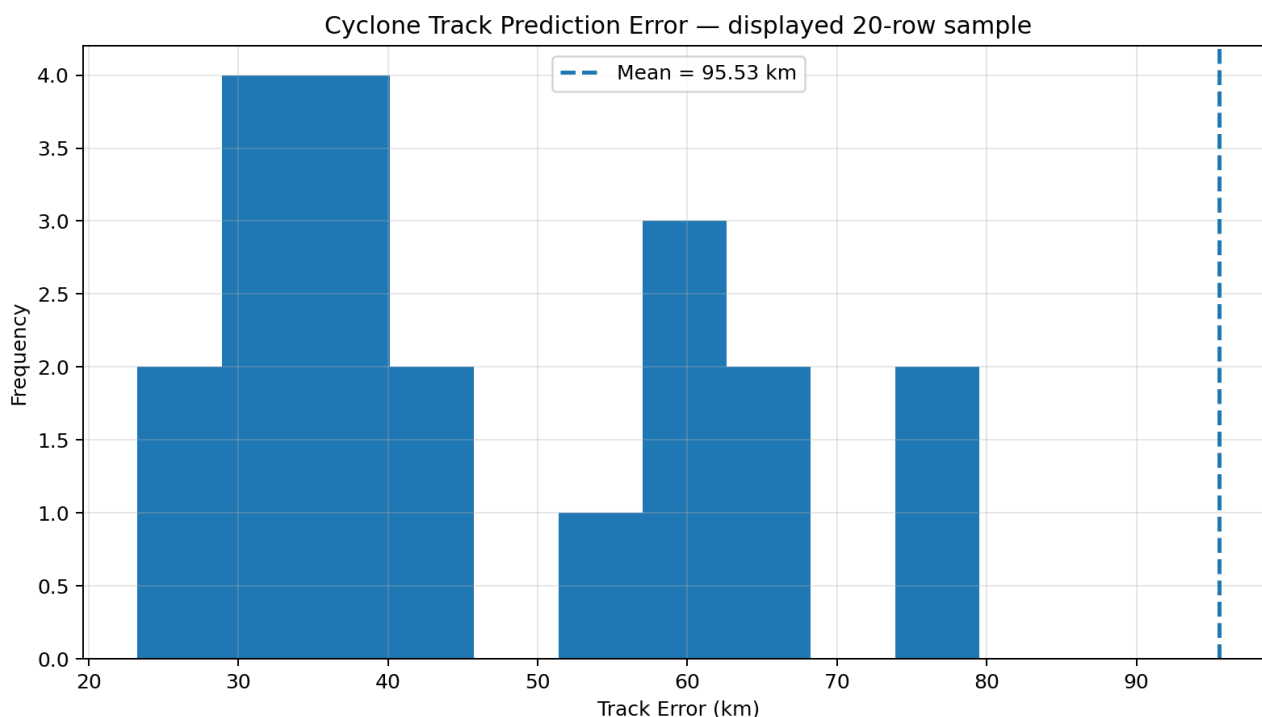


The final code generates these scatter plots for the full test set. Because the text export available for this report preserved only the first 20 displayed test rows rather than the complete prediction CSV, the reconstructed plots above use those 20 displayed rows and are explicitly labelled as such. The numerical test metrics in this report are the complete reported metrics.

9. Track Prediction Error

The model's predicted latitude and longitude are converted into a physical distance error using the Haversine formula.

Statistic	Reported value
Mean	95.53 km
Median	83.96 km
RMSE	113.12 km
Minimum	3.19 km
Maximum	501.59 km



The histogram is reconstructed from the 20 displayed prediction rows, while the mean/median/RMSE/min/max values above are the full test-set statistics reported by the final run. The reported mean 95.53 km should be presented as the **mean 6-hour track prediction error on the held-out test set**.

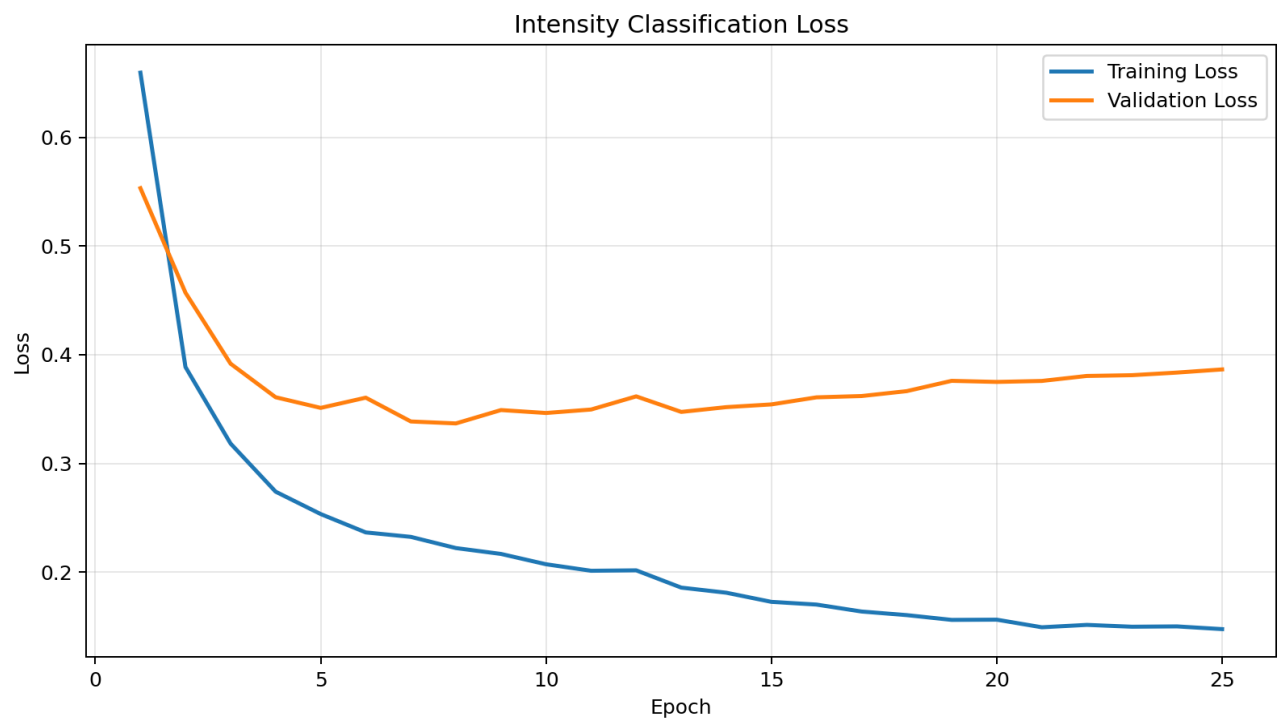
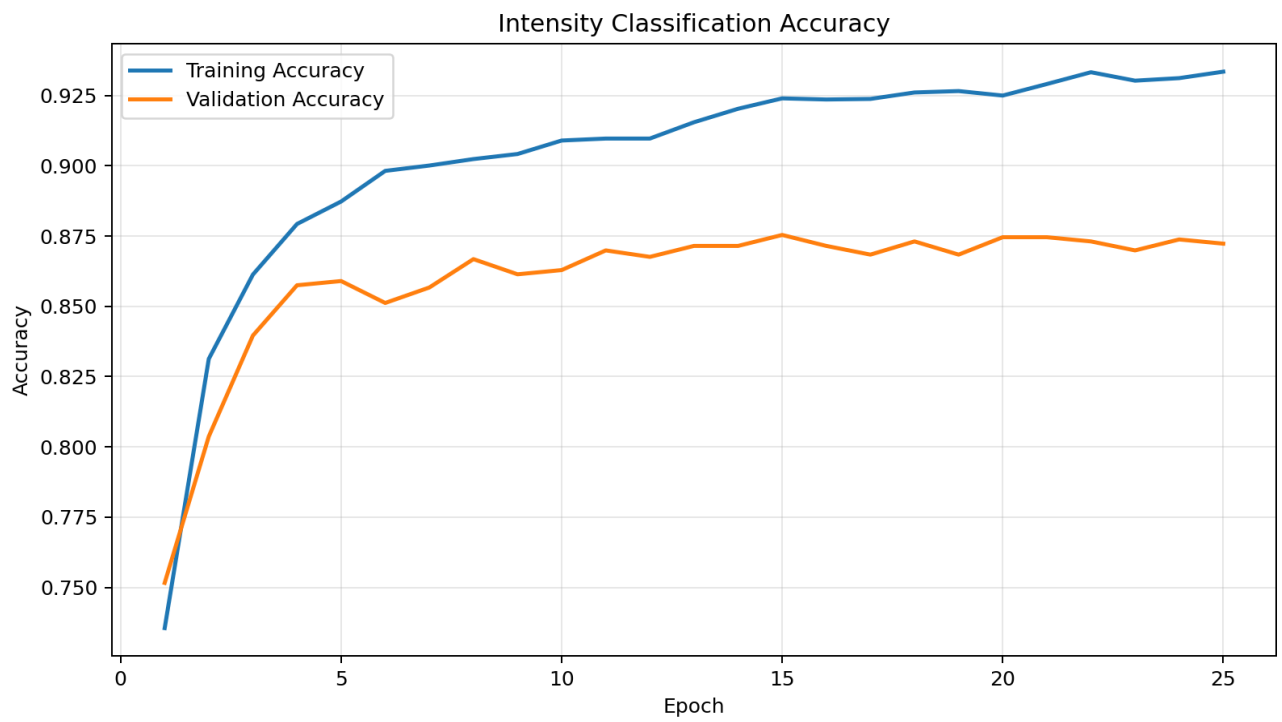
10. Intensity Classification

A second LSTM predicts one of three intensity categories. The classes are defined in the final pipeline as: **Weak** < 34 kt, **Moderate** = 34–63 kt, and **Strong** ≥ 64 kt. The classifier uses the same 6-step × 17-feature sequence structure.

Class	Training count
Weak	3,359
Moderate	1,654
Strong	655

Class weights were used to reduce the effect of class imbalance: approximately 0.562 for Weak, 1.142 for Moderate and 2.884 for Strong.

11. Classification Training Behaviour

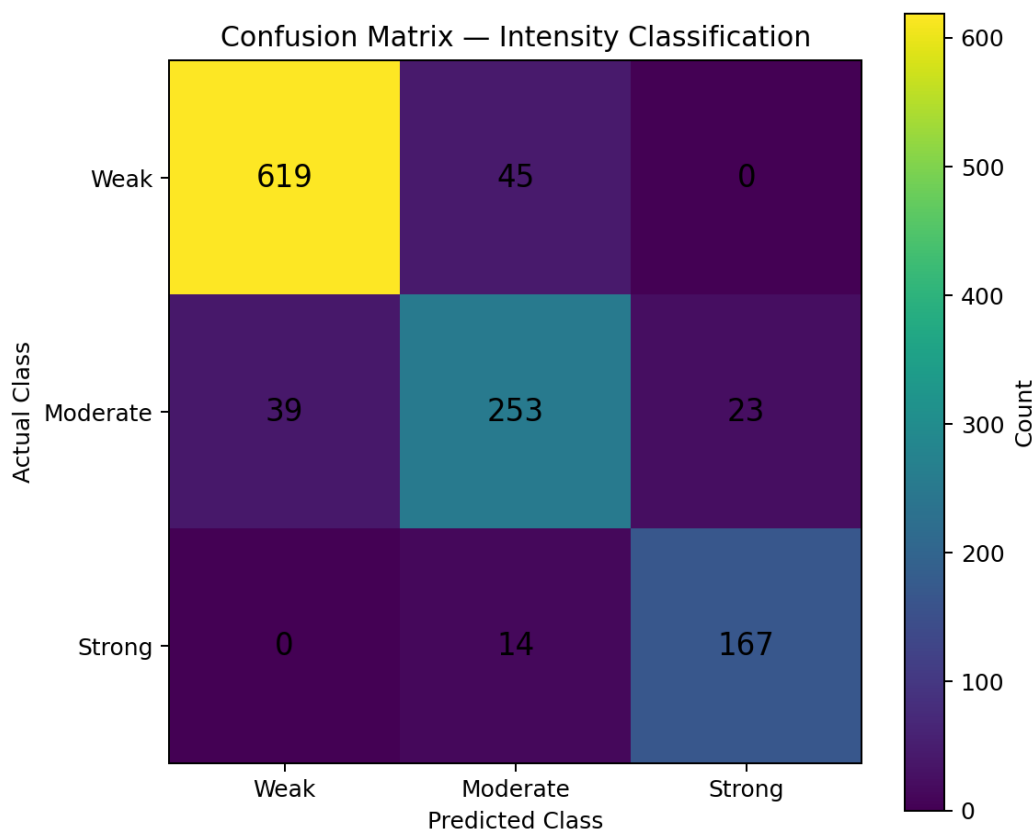


The best validation accuracy was 87.539% at epoch 15. Training continued to epoch 25 and then EarlyStopping restored epoch-15 weights.

12. Final Classification Results

Test accuracy: **89.57%** on 1,160 test sequences.

Class	Precision	Recall	F1	Support
Weak	0.94	0.93	0.94	664
Moderate	0.81	0.80	0.81	315
Strong	0.88	0.92	0.90	181



Confusion matrix: Weak → [619,45,0], Moderate → [39,253,23], Strong → [0,14,167]. The Strong class recall is 92%, meaning 167 of 181 strong test cases were correctly identified.

13. What the Model Can Tell the SIH System

From the final trained outputs, the system can provide a 6-hour-ahead estimate of cyclone location, wind speed and pressure, plus an intensity category. These outputs can support downstream warning and decision-support modules such as track visualisation, intensity alerts and risk communication. They should be presented as model predictions with uncertainty/validation context rather than as guaranteed outcomes.

Question	Model output
Where will it move?	Next 6-hour latitude + longitude
How strong will it be?	Next 6-hour wind speed + intensity class
What pressure is expected?	Next 6-hour central pressure
How accurate is the track?	Mean 6-hour track error = 95.53 km
How reliable is classification?	89.57% test accuracy

14. Training Outputs Saved in Kaggle

The final run saved the trained models and result tables under `/kaggle/working/`:

- `best_cyclone_regression.keras`
- `best_cyclone_intensity_classifier.keras`
- `cyclone_regression_test_results.csv`
- `cyclone_regression_metrics.csv`
- `cyclone_classification_results.csv`

15. Key Strengths of the Final Pipeline

- Cyclone-wise train/validation/test split reduces storm-level leakage.
- Scalers are fitted only on training data.
- Sequences require consecutive 3-hour observations and respect cyclone boundaries.
- Separate regression and classification objectives make the outputs easy to interpret.
- Checkpointing, learning-rate reduction and early stopping help control overfitting and preserve the best validation model.
- Evaluation includes both statistical metrics and a physical track-distance metric.

16. Important Scope / Reporting Note

The SIH problem statement calls for multi-source satellite data. The inspected final training code, however, builds the trained model from IBTrACS-derived track, intensity, motion, land-distance and time features. The Asian Typhoon files were inspected earlier but are not fused into this final trained LSTM. Therefore, for a technically accurate presentation, describe the current result as the **validated IBTrACS-based prediction module** within the broader Hexa-Minds multi-source cyclone system. A future fusion stage can add satellite observations and other meteorological sources to the model.

17. Final Takeaway

The final run completed successfully. The strongest reported result is the combination of **89.57% intensity classification accuracy** and **95.53 km mean 6-hour track error**, with very high R^2 values for the position targets and strong performance for wind and pressure. The model therefore demonstrates a solid proof-of-concept for short-term cyclone state prediction and intensity classification, while the next major SIH development step is true multi-source data fusion and, ideally, longer-horizon forecasting such as 24–72 hours.

Source basis: the user's final Kaggle training output and the corresponding final training code/notebook.